Anastasios Tsolakis

Robotics and Control Engineer
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Portfolio
LinkedIn
Scholar
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Profile

Robotics and Control Engineer (PhD, TU Delft) with expertise in dynamics, motion planning, fault-tolerant control, and ROS2-based system integration. Strong mechanical engineering foundation (kinematics, dynamics, mechanisms, design, CAD/CAE) and hands-on deployment of autonomous platforms (ASVs, UGVs). Solid background in dynamical systems, control, estimation, and optimization, with experience translating advanced research into validated engineering solutions. Collaborative team player with international, interdisciplinary experience and strong communication skills.

Experience

Research & Development (PhD) *Dec 2020 - Jun 2025*
TU Delft, The Netherlands

- Developed advanced motion planning algorithms for ASVs; robustness to faults/uncertainties and safer navigation.
- Integrated ROS-based planning and control pipelines in simulation; automated experiments and performance analysis.
- Collaborated in multidisciplinary teams and published in international venues and journals (IEEE T-ITS, ECC).
- Supervised students on their MSc thesis.

Teaching Assistant *Nov 2019 - Jan 2020*
TU Delft, The Netherlands

- Supported MSc-level course on nonlinear systems theory, guiding and evaluating students in assignments.

Mechanical Engineering Intern *Jul 2016 - Oct 2016*
Maniatopoulos S.A. IDEAL BIKES, Patras, Greece

- Designed and prototyped components and tools for a cargo e-bike.
- Performed statistical quality analysis to reduce frame defects and improve manufacturing robustness.

Formula Student Team Member *Jul 2013 - Oct 2014*
Aristotle Racing Team, Thessaloniki, Greece

- Led design, structural analysis (CAD/CAE), and manufacturing of the vehicle's steel tubular frame.
- Coordinated subsystem integration, subsystem interfaces, assembly sequencing, rule-compliance, and validation.
- Ensured competition readiness; achieved top-10 placements internationally.

Education

PhD in Cognitive Robotics *Dec 2020 – Jun 2025*
TU Delft, The Netherlands

- Thesis: *"Rule-compliant and Fault-tolerant Motion Planning: With Application to Autonomous Surface Vehicles"*.
- Developed real-time algorithms for collision avoidance with rule compliance, fault diagnosis, fault reconfiguration, and robustness to uncertainties, resulting in safer and more reliable navigation compared to existing solutions.
- Collaborated within an international, interdisciplinary research team and co-supervised MSc students on projects related to autonomous navigation.

MSc in Systems and Control *Sep 2018 – Dec 2020*
TU Delft, The Netherlands GPA: 8.56/10 ***Cum Laude***

- Thesis: *"Distributed IDA-PBC for Nonholonomic Mechanical Systems"*. Designed a distributed passivity-based control algorithm for a fleet of heterogeneous, underactuated, and nonholonomic robots.
- Specialized in control theory, dynamical systems, optimization, and adaptive/predictive/robust control methods.
- Collaborated in small international teams to deliver high-quality results in demanding course projects.

M.Eng. in Mechanical Engineering (Integrated BSc & MSc equivalent) *Sep 2011 – Jul 2018*
Aristotle University of Thessaloniki, Greece GPA: 8.08/10

- Thesis: *"The Effect of Power Distribution Architectures and Torque Vectoring in Vehicle Dynamics"*. Investigated vehicle dynamics under different power distribution architectures and developed a torque-vectoring controller.
- Specialized in mechanical design, analysis, synthesis, and control of mechanical systems.
- Member of Formula Student team; collaborated to deliver a competitive racing vehicle for international competitions.

Publications (Selected)

- **Tsolakis, A.**, Negenborn, R., Reppa, V., Ferranti, L. *Model Predictive Trajectory Optimization and Control for Autonomous Surface Vessels Considering Traffic Rules.* IEEE Trans. Intell. Transp. Syst., 25(8):9895–9908.
- **Tsolakis, A.**, Ferranti, L., Reppa, V. *Set-Membership Estimation for Fault Diagnosis of Nonlinear Systems.* In *Proc. European Control Conference (ECC)*, Thessaloniki, Greece, 24–27 Jun. 2025.
- **Tsolakis, A.**, Benders, D., De Groot, O., Negenborn, R., Reppa, V., Ferranti, L. *COLREGs-aware Trajectory Optimization for Autonomous Surface Vessels.* IFAC-PapersOnLine 55(31):269–274.
- **Tsolakis, A.**, Keviczky, T. *Distributed IDA-PBC for a Class of Nonholonomic Mechanical Systems.* IFAC-PapersOnLine 54(14):275–280.

Awards & Distinctions

- MSc Systems & Control *Cum Laude*, TU Delft.
- Young Author Award nominations (IFAC-MICNON 2021, IFAC-CAMS 2022).
- Formula Student: 4th overall (FS Italy); 12th overall (FS Germany); top-10 in Engineering Design (FS Hungary).

Skills

- **Programming:** C++, Python, Matlab/Simulink, ROS2
- **Tools:** Git; CAD (SolidWorks, Inventor, Creo Parametric); FEA/CAE (ANSA, Altair MotionView)
- **Languages:** Greek (native), English (fluent), French (A1, learning)
- **Interests:** Rock climbing, hiking, skiing, sailing, scuba diving, motorcycle trips, classical guitar

Home Robotics Project

- Designed a serviceable enclosure (mounting, cable management, access) and layout for onboard computers & sensors.
- Fused wheel odometry, IMU and 2D LiDAR in ROS2; performed calibration, time synchronization and state-estimation.
- Developing an MPC-based motion planner optimized for on-board efficiency and real-time constraints.
- Performed sensor/electrical bring-up, network configuration, logging, and analysis of field runs; basic operator UI.